Advantages of Printed Resistors

1. Low cost
2. It takes small space

Disadvantages

1. Replacement of faulty printed resistors is complicated and resulting in damage to the PCBs.

1.8.7 SMD Resistors

SMD stands for Surface Mounted Device. This type of resistor is directly soldered on copper solder points in the surface of the board and therefore they are called SMD resistors. They are smaller in size than their traditional carbon resistors. They are square, rectangular or oval in shape. The SMD resistor is shown in Figure 1.37 (a) and (b).

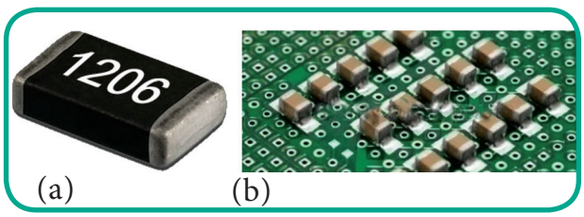


Figure 1.37 **SMD resistors**

Advantages

1. This type of resistor has small leads or points that are soldered on copper solder points in the surface of the board. This eliminates the need for holes in the PCB and we can utilize both sides of the PCB.
2. Once the PCB is manufactured, SMDs can be placed on it using a special machine called a pick and place machine.

Disadvantages

1. Replacement of faulty resistors is complicated.

1.8.8 Variable Resistor

The resistor which value can be changed is called as variable resistors.

Its symbol is shown in Figure 1.37 (a)

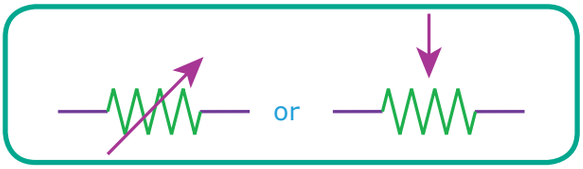


Figure 1.37(a) **Symbol of variable resistor**

Variable resistors are also called as potentiometers. In this, base is like a washer type with carbon coating. There is a rotary setup having slip-ring to the resistor and have three electrical contacts as shown in Figure 1.38. Between the first terminal T_1 and last terminal T_3, the value of resistance is fixed, i.e., cannot be altered. But centre terminal T_2 is a variable terminal with respect to T_1 and T_3.You can check this using a multimeter. A plastic or metal shaft attached to T_2 is used to change the value of the resistance.

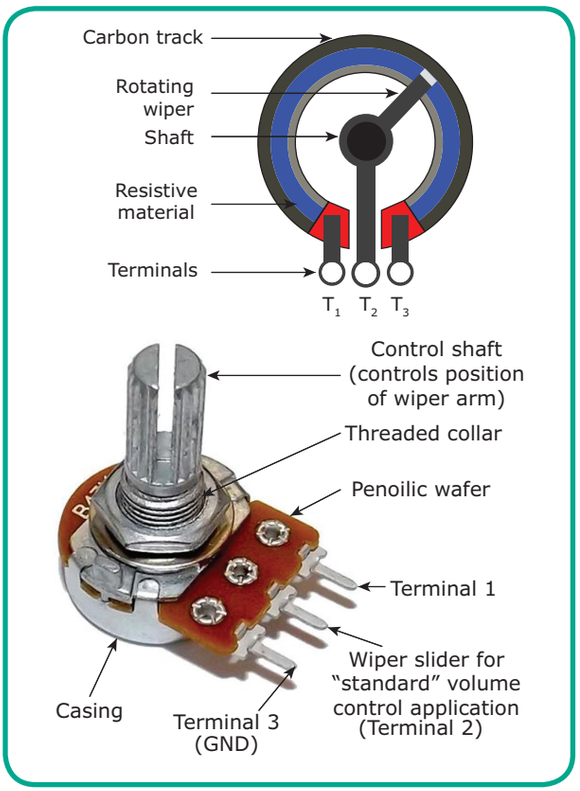


Figure 1.38 **Variable resistor**

ELEC_EQUIP_Chapter01.indd 19 11/10/2020 12:22:43 PM